

Technology Stack

Date: 30 September 2026
Team ID: SWTID-2026-3656
Project Name: EduGenie – Google Gemini Powered Learning Assistant
Maximum Marks: 5 Marks

Layer	Technology	Actual Use in EduGenie
Backend	Python + FastAPI	Application, routing, form handling, JSON responses
Server	Uvicorn	Runs the ASGI application locally/deployed
Templates	Jinja2	Renders <code>templates/index.html</code>
Frontend	HTML, CSS, JavaScript	Responsive animated learning dashboard
Cloud AI	Google Gemini via <code>google-genai</code>	Q&A, quiz, summarization, learning path
Local AI	Transformers + PyTorch	LaMini-Flan-T5 concept explanation
Tokenizer	SentencePiece	Supports the T5 tokenizer
Configuration	python-dotenv	Loads <code>.env</code> before AI modules read the key
Form parsing	python-multipart	Supports FastAPI form data

Environment

```
GEMINI_API_KEY=your_google_gemini_key
GEMINI_MODEL=gemini-2.5-flash
```